Normal subgroups and Factor Groups.

- For a subgroup H of a group G, the collection of left cosets of H in G, forms a group called the Factor group of G by H written as G/H. under the operation

$$(aH)(bH) = (ab)H$$

if and only if (theorem 10.4)

H is <u>normal</u> in G

meaning

$$\forall\, g \in G \qquad gH = Hg.$$

or equivalently (theorem 10.3)

$$\forall\, g \in G \qquad gHg^{-1} = H.$$

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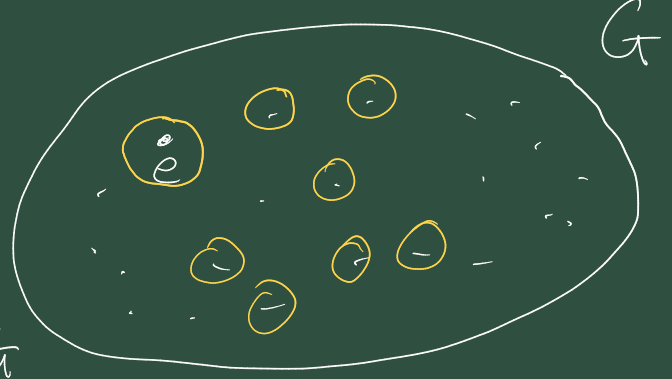
of course for any group G we can always point to two normal subgroups of G $\{e\}$, and G itself.

Look at the associated factor groups.

$G/\{e\}$



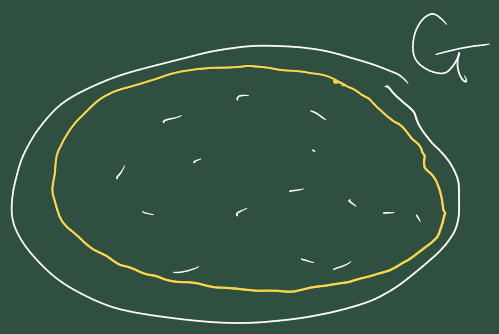
The cosets are all the singleton subsets $\{g\}$, $g \in G$.

$g\{e\}\, h\{e\} = gh\{e\}$.

So in fact $G/\{e\}$ just replicates exactly the group G and so

$$G/\{e\} \cong G.$$

What about G/G



$G/G = \{G\}$

its Cayley table is

	G
G	G

So G/G is the trivial group, consisting of just the idea.

A "<u>simple</u>" group is one having <u>no</u> non-trivial proper normal subgroups. (ie apart from $\{e\}$ and G itself).

So a simple group cannot be simplified in a non-trivial way by

factor group construction.

One important class of simple groups are the groups $\mathbb{Z}_p$, when p is prime (or equivalently, the cyclic groups of prime order). as these have no non-trivial proper subgroups at all.

In 10.2 its proved that the Alternating group A_n is simple for every $n \geq 5$.

We won't examine the proof of this

Exercises.

Q2 Find all subgroups of D_4. Which are normal? And examine the associated factor groups.

D_4 = gp of symmetries of square

under op. of composition.

$= \langle r, s \;\; r^4 = e, \; s^2 = e, \; sr = r^{-1}s \rangle.$

$$= \{ \underbrace{e, r, r^2, r^3}_{\text{rotations}}, \underbrace{s, rs, r^2s, r^3s}_{\text{reflections}} \}.$$

Subgroups of D_4

First non-interesting pair.

$\{e\}$

D_4

And the non-trivial proper subgroups are:

(thinking of sub-group conditions from Prop 3.30)

$R_1 = \langle r \rangle = \{e, r, r^2, r^3\}$. all conditions prop 3.30 satisfied here.

$S_1 = \{e, s\}$ " " "

$S_2 = \{e, rs\}$ " " "

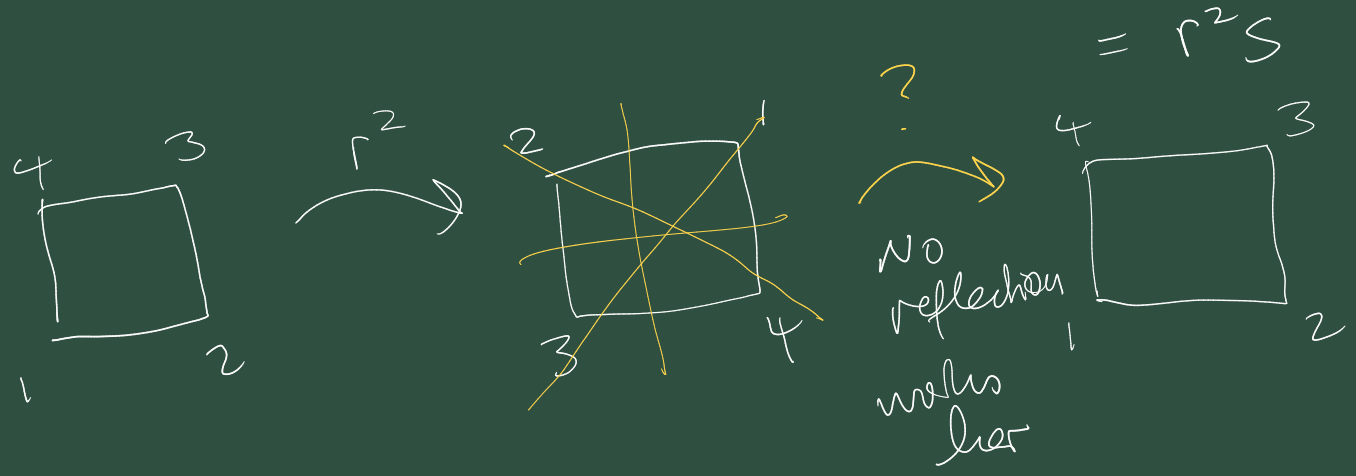
$S_3 = \{e, r^2s\}$ " " "

$S_4 = \{e, r^3s\}$ " " "

$R_2 = \{e, r^2\}$ " " "

$A = \{e, r^2, s, r^2s\}$. " " "

$$sr^2 = srr = r^{-1}sr = r^{-1}r^{-1}s = r^{-2}s$$
$$= (r^2)^{-1}s$$



$$\underline{s r^2 s} = r^2 s s = r^2 s^2 = r^2$$

A is a smaller dihedral group called D_2.

And the final subgroup

$B = \{ e, r^2, rs, r^3 s \}$. all conditions from Prop 3.30 satisfied.

No other subgroups.

Which of these six non-trivial proper subgroups are normal?

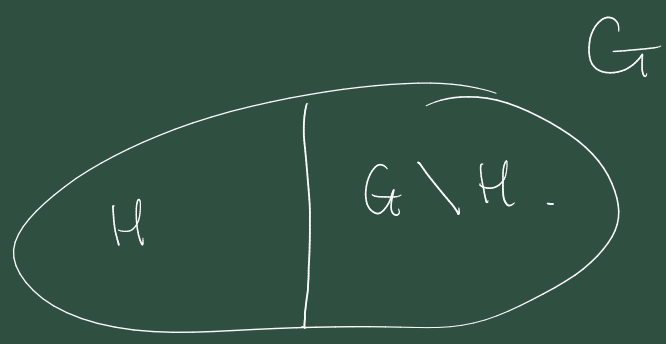
(Continue at 5:07.)

$R_1, R_2, S_1, S_2, S_3, S_4, A, B$.

Note $|R_1| = |A| = |B| = 4 = \frac{|D_4|}{2}$

and whenever a subgroup H of G has order $|H| = \frac{|G|}{2}$ it is automatically normal since there are only two left cosets.

H, $G \setminus H$



So therefore

R_1, A, B are ~~all~~ each normal in D_4.

Consider.

$S_1 = \{e, s\}$, rS_1r^{-1}

Using condition 2 of theorem 10.3.

Consider the conjugate element.

$$rsr^{-1} = r(r^{-1})^{-1}s$$
$$= rrs$$
$$= r^2 s \notin S_1.$$

So condition 2 is broken and S_1 is not normal in D_4.

Similarly we can show that ~~neither~~ all of S_2, S_3, S_4 are not normal in D_4.

Finally consider $R_2 = \{e, r^2\}$ and consider conjugates of r^2 by elements of D_4.

$$r^j r^2 r^{-j} = r^{j+2-j} \qquad j = 0,1,2,3$$
$$= r^2 \in R_2.$$

Now consider.

$$r^j s \underline{r^2} (r^j s)^{-1} \qquad j = 0,1,2,3$$
$$= r^j s r^2 r^j s$$
$$= r^j \underline{s r^{2+j}} s.$$
$$= r^j r^{-2-j} s s, \quad \text{using several } sr = r^{-1}s.$$
$$= r^j r^{-2-j} s^2,$$
$$= r^j r^{-2-j}, \quad s^2 = e.$$
$$= r^{j-2-j}$$
$$= r^{-2}$$
$$= (r^2)^{-1} = r^2 \in R_2.$$

So by cond 2 of theorem 10-3, R_2 is normal in D_4.

So R_1, A, B, R_2 are the four non-trivial proper normal subgroups of D_4.

For each of R_1, A, B, all of order 4. D_4/R_1, D_4/A, D_4/B are each

factor groups of order 2, and so isomorphic to $\mathbb{Z}_2$. $R_2 = \{e, r^2\}$

Finally.

$$D_4/R_2 = \{R_2, rR_2, sR_2, rsR_2\}$$

the four cosets.

its Cayley table is

	R_2	rR_2	sR_2	rsR_2
R_2	$\underline{R_2}$	rR_2	sR_2	rsR_2
rR_2	rR_2	$\underline{R_2}$	rsR_2	sR_2
sR_2	sR_2	rsR_2.	$\underline{R_2}$	rR_2.
rsR_2	rsR_2	sR_2	rR_2	$\underline{R_2}$.

this could be isomorphic to $\mathbb{Z}_4$, or $\mathbb{Z}_2 \times \mathbb{Z}_2$.

$R_2 = \{e, r^2\}$, $r^2 R_2 = \{r^2, r^4 = e\} = R_2$

$$sR_2rR_2 = srR_2$$
$$= \{sr, sr^3\}$$
$$= \{r^{-1}s, r^{-3}s\}$$
$$= \{r^3s, rs\} = rsR_2.$$

Note every non-id element of D_4/R_2 has order 2, so $D_4/R_2 \not\cong \mathbb{Z}_4$ as $\mathbb{Z}_4$ has elements of order 4, namely 1 and 3. So therefore

$$D_4/R_2 \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

<u>Q4</u> Consider the group T of matrices of the form $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$, a, c both non-zero. under matrix multiplication. (So in fact T is a subgroup of $GL(2, \mathbb{R})$.)

Define the subset U

$$U = \left\{ \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} : x \in \mathbb{R} \right\} \subset T.$$

a). Using Prop. 3.30.

clearly $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in U$, just use $x=0$.

Let $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & y \\ 0 & 1 \end{pmatrix} \in U$.

$$\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & y \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x+y \\ 0 & 1 \end{pmatrix} \in U.$$

$$\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -x \\ 0 & 1 \end{pmatrix} \in U$$

So by Prop. 3.30 U forms a subgroup of T.

b). U is abelian as

$$\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & y \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x+y \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & y+x \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & y \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$$

c). We will show, for any $\tau \in T$,

$\tau U \tau^{-1} \subset U$.

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \frac{1}{ac} \begin{pmatrix} c & -b \\ 0 & a \end{pmatrix}$$

$$\vdots$$

$$= \begin{pmatrix} 1 & \text{[illegible]} \\ 0 & 1 \end{pmatrix}$$

$\in U$.